

## A population proportion

A population proportion  $p$  is the fraction of population members with a specified outcome.

**What outcome would define  $p$  when studying use of a campus printing service?**

## A sample proportion

The sample proportion is the number of observed successes divided by the sample size.

$$\hat{p} = \frac{X}{n}$$

**For 128 successes among 400 observations, what is the sample proportion?**

## Two notation choices

This textbook uses  $p'$  for the sample proportion;  $\hat{p}$  denotes the same statistic.

$$p' = \hat{p}$$

**Which symbol refers to the unknown population proportion?**

## The failure proportion

For population proportions,  $q = 1 - p$ . For sample proportions,  $q' = 1 - p'$ , also written  $1 - \hat{p}$ .

**If 128 of 400 observations are successes, what is  $q'$ ?**

## Count or proportion?

Synthetic random sample: 128 of 400 students used the printing service this week.

**Which number is a success count, and which number is the sample proportion?**

## A binary outcome

The binomial distribution counts successes in  $n$  independent binary trials with constant probability  $p$ . The sample proportion divides this count by  $n$ .

**Would average pages printed per student be a proportion problem?**

## Sampling variability of proportions

For large enough expected success and failure counts, the sample proportion is approximately normal with mean  $p$  and SD  $\sqrt{p(1 - p)/n}$ .

$$SD(\hat{p}) = \sqrt{\frac{p(1 - p)}{n}}$$

**What happens to this SD when  $n$  quadruples?**

## A proportion sampling model

Independent binary observations have  $p = 0.4$  and  $n = 100$ . Both expected counts exceed five.

**What normal distribution approximates the sample proportion?**



## Standardizing a sample proportion

Use the population  $p$  to describe sampling variation when  $p$  is known.

$$z = \frac{p' - p}{\sqrt{p(1 - p)/n}}$$

**For  $p = 0.4$ ,  $n = 100$ , and  $p' = 0.5$ , what z-score results?**

## An estimated standard error

A proportion interval estimates the standard error by replacing unknown  $p$  with the observed sample proportion.

$$\widehat{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

**For  $\hat{p}=0.32$  and  $n=400$ , what is the estimated standard error?**

## Successes and failures

The textbook requires more than five observed successes and more than five observed failures for its ordinary normal-approximation proportion interval.

**Does a synthetic sample with 128 successes and 272 failures pass this check?**

## A sparse success sample

Synthetic random sample: 2 successes and 38 failures.

**Would a normal-approximation proportion interval meet the success-failure check?**

## Independence for proportions

Random sampling supports representativeness; independence also matters for the standard error formula.

**Would surveying every member of a few randomly selected clubs automatically give independent responses?**

## A sample fraction

For sampling without replacement,  $n \leq 10\%$  of the population is a common independence approximation.

**Would a sample of 400 from 20,000 students meet that guideline?**

## A proportion margin of error

The error bound for a population proportion (EBP) is a critical z value times the estimated standard error.

$$EBP = z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

**For  $\hat{p}=0.32$ ,  $n=400$ , and  $z^*=1.96$ , what is EBP?**

## Constructing a proportion interval

Synthetic independent random sample: 128 of 400 students used printing this week. The population is large; use 95% confidence and  $z^*=1.96$ .

**What is the normal-approximation confidence interval for  $p$ ?**



## Interpreting a proportion interval

Synthetic result: a 95% interval for printing-service use is 0.274 to 0.366.

**How would you state this interval in percentages for the population?**

## Percentage points

Synthetic result: an estimated proportion is 32%, with margin of error 4.6 percentage points.

### What are the percentage endpoints?

## Confidence is about a procedure

A valid 95% proportion interval procedure captures the fixed  $p$  in about 95% of repeated intervals.

**Does 95% mean that 95% of students use the printing service?**

## The plus-four adjustment

A plus-four interval uses two added successes and two added failures in its adjusted proportion.

$$\tilde{p} = \frac{x + 2}{n + 4}$$

**For a synthetic sample with 8 successes among 40 observations, what is the adjusted proportion?**

## When to use plus four

The textbook recommends plus four when confidence is at least 90% and the sample size is at least 10.

**Would a sample of 24 observations at 85% confidence meet these recommendations?**

## A plus-four interval

Synthetic independent random sample: 8 of 40 students used printing this week. Large population; 95% confidence;  $z^*=1.96$ .

$$\tilde{p} \pm z^* \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{n + 4}}$$

**What plus-four confidence interval for p follows?**

## Changing confidence

Use the same observed sample proportion and sample size. Compare 90% and 99% confidence intervals.

**Which margin of error is larger?**

## Changing sample size

Hold the observed proportion and confidence level fixed. Increase  $n$  from 400 to 1,600.

**How does the estimated margin of error change?**



## Planning a proportion sample

A planning estimate  $p_0$  gives a sample-size formula, rounded upward, for a desired margin  $E$ .

$$n = \frac{(z^*)^2 p_0 (1 - p_0)}{E^2}$$

**For  $p_0=0.32$ ,  $E=0.03$ , and  $z^*=1.96$ , what  $n$  is needed?**

## Planning without an estimate

When no planning estimate is available,  $p_0=0.5$  gives the largest value of  $p_0(1-p_0)$ .

**Why would this choice be conservative for planning sample size?**

## **A conservative planning problem**

Plan a 95% proportion interval with margin at most 0.03. No prior proportion estimate is available; use  $z^*=1.96$ .

**What sample size should be planned?**

## **Precision cannot remove bias**

Synthetic online poll: 90% of 5,000 voluntary respondents favor longer library hours.

**Would its small calculated margin of error establish representativeness of all students?**

## A claim and an interval

Synthetic result: a 95% interval for printing-service use is 0.274 to 0.366. A claim states  $p=0.50$ .

**Is the claimed proportion compatible with this interval?**

## Estimate versus null model

For a proportion interval, standard error uses  $\hat{p}$ . A proportion test may instead use a hypothesized  $p_0$ .

**Why could two analyses of the same sample use different standard errors?**

## Sketching the confidence interval

Synthetic survey: a 95% interval for  $p$  is (0.42, 0.58), with sample proportion 0.50.

**How would you mark the estimate, endpoints, and error bound, and show the critical tail areas on a z curve?**

## Recovering a proportion estimate

A synthetic survey reports a 95% confidence interval from 0.36 to 0.44.

**What are its point estimate and error bound in proportion units and percentage points?**